

Construction π_A over Multiquadratic Fields for Compound Block-Fading Wiretap Channels

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Abstract

We construct multilevel lattice codes from multiquadratic number fields for the compound block-fading wiretap channel. More precisely, we specialize Construction π_A over the ring of integers $\mathcal{O}_K$ and exploit rational primes that split completely in K to obtain a Chinese Remainder Theorem (CRT) decomposition into small residue alphabets, notably binary, which enables multistage decoding. The resulting nested lattices fit into the algebraic Construction A framework and, when combined with discrete Gaussian shaping and flatness-factor bounds, provide universal reliability for the legitimate receiver and strong secrecy uniformly over the eavesdropper compound set.

Key Words : Multiquadratic number fields, Lattice codes, Construction π_A lattice, Wiretap channel.

1 Introduction

Lattice codes are finite constellations obtained by intersecting a lattice with a bounded shaping region. They can approach the capacity of the additive white Gaussian noise (AWGN) channel and support algebraic operations such as compute-and-forward and distributed source coding [NCL10, KP07, Zam14].

Lattices also provide a natural framework for physical-layer security. In a wiretap channel, a legitimate receiver and an eavesdropper observe noisy

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versions of the same transmission [Wyn75, LH78]. Nested lattice codes combined with discrete Gaussian shaping and flatness-factor analysis can achieve strong secrecy [LLBS14]. This approach has been extended to algebraic Construction A lattices over number fields and to compound multiple-input multiple-output (MIMO) models, yielding universal reliability and flatness-factor-based secrecy guarantees [CLB18, CLB20].

From a practical viewpoint, multilevel lattice constructions such as Construction π_A are appealing because they combine small-alphabet component codes through the CRT and admit multistage decoding, thereby reducing decoding complexity [HN17]. Variants over number fields and quaternionic orders have shown good reliability [HN17, HHVN19, SCL25]. However, to the best of our knowledge, the physical-layer security of Construction π_A for compound block-fading wiretap channels has not yet been studied.

In this work, we adapt Construction π_A to the ring of integers $\mathcal{O}_K$ of multiquadratic number fields K , focusing on rational primes that split completely. This produces a CRT decomposition into small residue alphabets, notably binary ones, and hence enables multistage decoding. We illustrate the construction on $K = \mathbb{Q}(\sqrt{17}, \sqrt{33})$, where $\langle 2 \rangle$ splits into four coprime ideals. The resulting nested lattice pairs fit into the algebraic Construction A framework of [CLB18, CLB20], which allows us to obtain universal reliability for Bob and strong secrecy for the corresponding compound block-fading wiretap channel.

Section 2 recalls the algebraic preliminaries on multiquadratic number fields. Section 3 develops Construction π_A over $\mathcal{O}_K$ and presents a binary example over $K = \mathbb{Q}(\sqrt{17}, \sqrt{33})$. Section 4 applies the resulting nested lattices to a compound block-fading wiretap channel and includes a finite-length simulation. Finally, Section 5 concludes.

2 Preliminaries

We recall the algebraic number theory ingredients needed for our lattice construction; see, e.g., [Mar77, Jar14].

A number field K is a finite extension of $\mathbb{Q}$ of degree $m = [K : \mathbb{Q}]$. Its ring of integers $\mathcal{O}_K$ is a free $\mathbb{Z}$ -module of rank m . For an ideal $\mathfrak{a} \subseteq \mathcal{O}_K$, its norm is $N(\mathfrak{a}) = |\mathcal{O}_K/\mathfrak{a}|$.

The CRT is fundamental to our construction: if $\mathfrak{p}_1, \dots, \mathfrak{p}_r$ are pairwise coprime ideals and $\mathfrak{P} = \prod_{j=1}^r \mathfrak{p}_j$, then

$$\mathcal{O}_K/\mathfrak{P} \cong (\mathcal{O}_K/\mathfrak{p}_1) \times \cdots \times (\mathcal{O}_K/\mathfrak{p}_r).$$

Definition 1. *A biquadratic field is a field of the form $K = \mathbb{Q}(\sqrt{a}, \sqrt{b})$, where a, b are distinct square-free integers. Then $[K : \mathbb{Q}] = 4$, and K contains the three quadratic subfields $\mathbb{Q}(\sqrt{a})$, $\mathbb{Q}(\sqrt{b})$, and $\mathbb{Q}(\sqrt{k})$, where $k = ab/\gcd(a, b)^2$.*

For a biquadratic field $K = \mathbb{Q}(\sqrt{a}, \sqrt{b})$, the ring of integers $\mathcal{O}_K$ admits an explicit integral basis depending on the congruence classes of a, b , and k modulo 4; see [Jar14].

A rational prime p factors in $\mathcal{O}_K$ as $p\mathcal{O}_K = \prod_{i=1}^r \mathfrak{p}_i^{e_i}$, where $f_i = [\mathcal{O}_K : \mathbb{F}_p]$ and $\sum_i e_i f_i = m$. We say that p splits completely if $r = m$ and $e_i = f_i = 1$ for all i .

For a quadratic field $\mathbb{Q}(\sqrt{a})$ and an odd prime $p \nmid a$, the Legendre symbol determines the splitting behaviour:

$$\left(\frac{a}{p}\right) = 1 \Rightarrow p \text{ splits}, \quad \left(\frac{a}{p}\right) = -1 \Rightarrow p \text{ is inert}.$$

If $p \mid a$, then p ramifies. For $p = 2$, the usual congruence criteria apply.

Proposition 1 ([Mar77]). *If p splits completely in both $\mathbb{Q}(\sqrt{a})$ and $\mathbb{Q}(\sqrt{b})$, then it splits completely in $\mathbb{Q}(\sqrt{a}, \sqrt{b})$.*

Therefore, when a rational prime splits completely in a biquadratic field, the CRT yields a decomposition into several small residue fields. This is the key algebraic ingredient used in the next section to define Construction π_A over $\mathcal{O}_K$ and to enable multistage decoding.

3 Lattice Constructions from Multiquadratic Fields

In this section, we recall algebraic lattices via the canonical embedding and define Construction π_A over $\mathcal{O}_K$ using the CRT, leading to nested lattice pairs with explicit design rate.

3.1 Algebraic Lattices and Canonical Embeddings

A lattice in $\mathbb{R}^N$ is a discrete subgroup generated by N linearly independent vectors. If $G = [\mathbf{v}_1 | \cdots | \mathbf{v}_N]$ is a generator matrix, then

$$\Lambda = \{G\mathbf{u} : \mathbf{u} \in \mathbb{Z}^N\}.$$

Let K be a number field of degree $m = [K : \mathbb{Q}]$ and signature (r_1, r_2) . The canonical embedding $\sigma_K : K \hookrightarrow \mathbb{R}^{r_1} \times \mathbb{C}^{r_2} \simeq \mathbb{R}^m$ maps $\mathcal{O}_K$ to a full-rank lattice in $\mathbb{R}^m$. In the biquadratic case $K = \mathbb{Q}(\sqrt{a}, \sqrt{b})$, we have $m = 4$; in particular, if $a, b > 0$, then K is totally real and $(r_1, r_2) = (4, 0)$.

More generally, if $\{x_1, \dots, x_m\}$ is a $\mathbb{Z}$ -basis of a $\mathbb{Z}$ -module $M \subset K$, then

$$\sigma_K(M) = \left\{ \sum_{i=1}^m u_i \sigma_K(x_i) \mid u_i \in \mathbb{Z} \right\}$$

is a lattice in $\mathbb{R}^m$.

In our setting, lattice points lie in $\mathcal{O}_K^n$, where n denotes the code length over $\mathcal{O}_K$. Applying the canonical embedding componentwise, we obtain Euclidean lattices in $\mathbb{R}^N$, where $N = mn$. By abuse of notation, we still denote this componentwise embedding by $\sigma_K : \mathcal{O}_K^n \hookrightarrow \mathbb{R}^N$.

The volume of the lattice associated with $\mathcal{O}_K$ is $\text{vol}(\sigma_K(\mathcal{O}_K)) = 2^{-r_2} \sqrt{|d_K|}$, where d_K denotes the discriminant of K . More generally, for an ideal $\mathfrak{a} \subseteq \mathcal{O}_K$, $\text{vol}(\sigma_K(\mathfrak{a})) = 2^{-r_2} \sqrt{|d_K|} N(\mathfrak{a})$, [Sam13].

3.2 Construction π_A Lattices

We now adapt Construction A to a multilevel setting using the CRT, following Construction π_A in [HN17].¹

Let $\mathfrak{p}_1, \dots, \mathfrak{p}_r \subset \mathcal{O}_K$ be pairwise coprime prime ideals, and set $\mathfrak{P} = \prod_{j=1}^r \mathfrak{p}_j$. Let $\rho : \mathcal{O}_K^n \rightarrow (\mathcal{O}_K/\mathfrak{P})^n$ denote the componentwise reduction modulo $\mathfrak{P}$, and let $\phi : \mathcal{O}_K/\mathfrak{P} \rightarrow \mathcal{O}_K/\mathfrak{p}_1 \times \dots \times \mathcal{O}_K/\mathfrak{p}_r$ be the CRT isomorphism.

Definition 2 (Construction π_A lattice). *Let $\mathcal{C}_j \subseteq (\mathcal{O}_K/\mathfrak{p}_j)^n$ be (n, k_j) -linear codes for $j = 1, \dots, r$, and define*

$$\mathcal{C} := \phi^{-1}(\mathcal{C}_1 \times \dots \times \mathcal{C}_r) \subseteq (\mathcal{O}_K/\mathfrak{P})^n.$$

The corresponding Construction π_A lattice is $\Lambda_{\pi_A}(\mathcal{C}) := \rho^{-1}(\mathcal{C}) \subseteq \mathcal{O}_K^n$. Equivalently, $\Lambda_{\pi_A}(\mathcal{C}) = \mathcal{C} + \mathfrak{P}^n$.

This construction enables multistage decoding by decomposing the lattice decoding into smaller component decoding tasks over the residue fields $\mathcal{O}_K/\mathfrak{p}_j$. Its volume is $\text{vol}(\Lambda_{\pi_A}(\mathcal{C})) = \prod_{j=1}^r N(\mathfrak{p}_j)^{n-k_j} 2^{-nr_2} |d_K|^{n/2}$.

3.2.1 Binary Example

We illustrate the construction when the rational prime 2 splits completely, so that the residue alphabets are binary.

First consider $K = \mathbb{Q}(\sqrt{17})$. Writing $\omega = \frac{1+\sqrt{17}}{2}$, the ideal $\langle 2 \rangle$ splits as $\langle 2 \rangle = \mathfrak{p}_1 \mathfrak{p}_2$. Hence, by the CRT, $\mathcal{O}_K/\langle 2 \rangle \cong \mathcal{O}_K/\mathfrak{p}_1 \times \mathcal{O}_K/\mathfrak{p}_2 \cong \mathbb{F}_2 \times \mathbb{F}_2$. A convenient isomorphism is

$$\begin{aligned} \Psi : \mathcal{O}_K/\langle 2 \rangle &\rightarrow \mathbb{F}_2 \times \mathbb{F}_2, \\ \alpha &\mapsto (\alpha \bmod \mathfrak{p}_1, \alpha \bmod \mathfrak{p}_2). \end{aligned}$$

Now consider the biquadratic field $K = \mathbb{Q}(\sqrt{17}, \sqrt{33})$. Since $\langle 2 \rangle$ splits completely in both $\mathbb{Q}(\sqrt{17})$ and $\mathbb{Q}(\sqrt{33})$, Proposition 1 implies that it also splits completely in K : $\langle 2 \rangle = \mathfrak{p}_1 \mathfrak{p}_2 \mathfrak{p}_3 \mathfrak{p}_4$. The explicit prime ideals can be computed in SageMath, [Dev24]. Therefore,

$$\mathcal{O}_K/\langle 2 \rangle \cong \mathcal{O}_K/\mathfrak{p}_1 \times \dots \times \mathcal{O}_K/\mathfrak{p}_4 \cong \mathbb{F}_2^4.$$

¹For a linear code $C \subseteq \mathbb{Z}_p^N$, Construction A defines the lattice $\Lambda_A(C) = \{x \in \mathbb{Z}^N : x \bmod p \in C\}$. In the algebraic setting, $\mathbb{Z}$ and $\mathbb{Z}_p$ are replaced by $\mathcal{O}_K$ and $\mathcal{O}_K/\mathfrak{p}$.

Thus Construction π_A over K can be built from four binary component codes, substantially reducing decoding complexity while preserving the algebraic structure needed for nested lattice coding.

Remark 1. *Construction π_A admits multistage decoding strategies such as the serial modulo decoder (SMD) and the parallel modulo decoder (PMD) from [HN17]. SMD decodes the CRT levels successively, whereas PMD decodes them in parallel. Both have comparable arithmetic complexity per code-word, while PMD may offer lower latency on parallel hardware at a modest performance loss.*

3.3 Nested Lattice Codes from Construction π_A over Multi-quadratic Fields

We now specialize Construction π_A to build nested lattice pairs for the wiretap setting. Let K be a multiquadratic field of degree $m = [K : \mathbb{Q}]$ with ring of integers $\mathcal{O}_K$. Let $\mathfrak{p}_1, \dots, \mathfrak{p}_r \subset \mathcal{O}_K$ be pairwise coprime ideals, with $q_j = N(\mathfrak{p}_j)$, $\mathfrak{P} = \prod_{j=1}^r \mathfrak{p}_j$.

For each level $j \in \{1, \dots, r\}$, choose nested linear codes $\mathcal{C}_e^{(j)} \subseteq \mathcal{C}_b^{(j)} \subseteq (\mathcal{O}_K/\mathfrak{p}_j)^n$, with dimensions $k_{e,j} \leq k_{b,j}$. Using the CRT isomorphism from Section 3.2, define the global nested codes

$$\begin{aligned}\mathcal{C}_e &:= \phi^{-1}(\mathcal{C}_e^{(1)} \times \dots \times \mathcal{C}_e^{(r)}) \subseteq (\mathcal{O}_K/\mathfrak{P})^n, \\ \mathcal{C}_b &:= \phi^{-1}(\mathcal{C}_b^{(1)} \times \dots \times \mathcal{C}_b^{(r)}) \subseteq (\mathcal{O}_K/\mathfrak{P})^n.\end{aligned}$$

Applying the Construction π_A lifting from Definition 2, we obtain

$$\tilde{\Lambda}_e = \mathcal{C}_e + \mathfrak{P}^n, \quad \tilde{\Lambda}_b = \mathcal{C}_b + \mathfrak{P}^n,$$

with $\tilde{\Lambda}_e \subseteq \tilde{\Lambda}_b \subseteq \mathcal{O}_K^n$.

Using the canonical embedding $\sigma_K : \mathcal{O}_K^n \hookrightarrow \mathbb{R}^N$ and a scaling factor $\gamma > 0$, we obtain the Euclidean nested lattice pair

$$\Lambda_e := \gamma \sigma_K(\tilde{\Lambda}_e), \quad \Lambda_b := \gamma \sigma_K(\tilde{\Lambda}_b),$$

with $\Lambda_e \subseteq \Lambda_b \subset \mathbb{R}^N$. Here Λ_b will serve as Bob's fine lattice, whereas Λ_e will be the coarse lattice used for shaping and secrecy.

The quotient size is $|\Lambda_b/\Lambda_e| = \frac{\text{vol}(\Lambda_e)}{\text{vol}(\Lambda_b)} = \prod_{j=1}^r q_j^{k_{b,j} - k_{e,j}}$, and therefore the design rate is $R_{\text{design}} = \frac{1}{N} \log_2 |\Lambda_b/\Lambda_e| = \frac{1}{mn} \sum_{j=1}^r (k_{b,j} - k_{e,j}) \log_2 q_j$. The scaling γ is chosen to satisfy the transmit power constraint.

Remark 2. *The simplest multistage decoding arises when the ideals $\mathfrak{p}_j$ lie above rational primes that split completely in K . In that case, $\mathcal{O}_K/\mathfrak{p}_j \cong \mathbb{F}_{p_j}$, so each level is defined over a small finite field. More generally, the construction applies to any collection of pairwise coprime ideals, with residue field sizes $q_j = N(\mathfrak{p}_j)$ not necessarily prime.*

4 Wiretap Application: Compound Block Fading

We now apply the nested lattices from multiquadratic Construction π_A to a compound block-fading wiretap channel. The nested pair $\Lambda_e \subset \Lambda_b \subset \mathbb{R}^N$ plays two distinct roles: the fine lattice Λ_b is used to guarantee reliable decoding at Bob, whereas the coarse lattice Λ_e is used for shaping and for hiding the transmitted coset from Eve. Our approach combines the universality result of [CLB18] with the flatness-factor based secrecy framework of [LLBS14, CLB20].

4.1 Channel Model and Compound Sets

As in Section 3.3, let $\Lambda_e \subset \Lambda_b \subset \mathbb{R}^N$ be a nested lattice pair obtained from multiquadratic Construction π_A . In this section, rates are measured in nats per channel use.

We consider a quasi-static compound block-fading complex wiretap channel with n_a transmit antennas, n_b antennas at Bob, and n_e antennas at Eve. During one codeword of T channel uses,

$$Y_b = H_b X + Z_b, \quad Y_e = H_e X + Z_e,$$

where $H_b \in \mathbb{C}^{n_b \times n_a}$ and $H_e \in \mathbb{C}^{n_e \times n_a}$ are the channel matrices, $X \in \mathbb{C}^{n_a \times T}$ is the transmitted codeword, and Z_b, Z_e have i.i.d. entries distributed as $\mathcal{CN}(0, \sigma_b^2)$ and $\mathcal{CN}(0, \sigma_e^2)$, respectively. We impose the average power constraint $\frac{1}{n_a T} \mathbb{E}[\|X\|_F^2] \leq P$, and define $\text{SNR}_b = \frac{P}{\sigma_b^2}$ and $\text{SNR}_e = \frac{P}{\sigma_e^2}$.

Following [CLB20], we define the admissible channel families

$$S_b := \left\{ H \in \mathbb{C}^{n_b \times n_a} : |I_{n_a} + \text{SNR}_b H^\dagger H| \geq e^{C_b} \right\}, \quad (1)$$

$$S_e := \left\{ H \in \mathbb{C}^{n_e \times n_a} : |I_{n_a} + \text{SNR}_e H^\dagger H| \leq e^{C_e} \right\}. \quad (2)$$

A channel realization is admissible if $H_b \in S_b$ and $H_e \in S_e$. Thus S_b collects the channel realizations for which Bob's channel is sufficiently good, while S_e collects those for which Eve's channel is no stronger than the prescribed threshold. The corresponding secrecy-capacity lower bound is $C_s = (C_b - C_e)^+$, and our goal is to achieve any secrecy rate $R < C_s$ uniformly over all admissible pairs $(H_b, H_e) \in S_b \times S_e$ (up to the constant gap inherited from the reduction step).

Vectorizing the channel gives

$$\text{vec}(Y_b) = (I_T \otimes H_b) \text{vec}(X) + \text{vec}(Z_b), \quad \text{vec}(Y_e) = (I_T \otimes H_e) \text{vec}(X) + \text{vec}(Z_e).$$

After passing to the equivalent real model induced by the lattice embedding, we obtain $\mathbf{y}_b = H_b^{\text{eq}} \mathbf{x}_R + \mathbf{z}_b$, and $\mathbf{y}_e = H_e^{\text{eq}} \mathbf{x}_R + \mathbf{z}_e$, where $\mathbf{x}_R \in \mathbb{R}^N$ corresponds to a point in Λ_b .

4.2 Reliability

We now show that the fine lattices from multiquadratic Construction π_A are universally good for Bob's compound channel family S_b , meaning that they remain decodable for every admissible realization in S_b . The key observation is that our CRT-based multilevel construction is a sub-ensemble of the algebraic Construction A lattices over $\mathcal{O}_K$ studied in [CLB18, CLB20].

Fix a code length n and pairwise coprime prime ideals $\mathfrak{p}_1, \dots, \mathfrak{p}_r \subset \mathcal{O}_K$, and set $\mathfrak{P} := \prod_{j=1}^r \mathfrak{p}_j$. At each level j , choose a linear code $\mathcal{C}_j \subset (\mathcal{O}_K/\mathfrak{p}_j)^n$ from a balanced ensemble in the sense of [Cam18], and define the CRT code $\mathcal{C}_b := \phi^{-1}(\mathcal{C}_1 \times \dots \times \mathcal{C}_r) \subset (\mathcal{O}_K/\mathfrak{P})^n$. The corresponding fine lattice Λ_b is therefore an algebraic Construction A lattice over $\mathcal{O}_K$ with modulus $\mathfrak{P}$.

Let $\mathbb{L}_n$ denote the ensemble obtained by drawing the component codes independently from balanced ensembles and forming the corresponding lattices Λ_b . By [CL20, SCL25], Construction π_A yields a balanced ensemble, and hence inherits the same averaging properties needed in [CLB18].

Proposition 2 (Minkowski–Hlawka and compactification). *For each n , the multiquadratic Construction π_A ensemble $\mathbb{L}_n$ satisfies: (i) (Minkowski–Hlawka) For every non-negative continuous function f with compact support,*

$$\mathbb{E}_{\Lambda_b \in \mathbb{L}_n} \left[\sum_{\lambda \in \Lambda_b \setminus \{0\}} f(\lambda) \right] \leq \frac{1}{\text{vol}(\Lambda_b)} \int_{\mathbb{R}^N} f(\mathbf{u}) d\mathbf{u} + \varepsilon_n,$$

with $\varepsilon_n \rightarrow 0$ as $n \rightarrow \infty$. (ii) (Compactification) The family $\{\mathbb{L}_n\}_{n \geq 1}$ compactifies the normalized block-fading channel family in the sense of [CLB18]: for every normalized channel matrix $\tilde{H}$ with $\det(\tilde{H}^\dagger \tilde{H}) = 1$, one can write $\tilde{H} = E_{\tilde{H}} U_{\tilde{H}}$, where $\|E_{\tilde{H}}\|$ is uniformly bounded and $U_{\tilde{H}}$ preserves the distribution of the ensemble.

Sketch of proof. Item (i) follows from the balancedness of the CRT code ensemble over $(\mathcal{O}_K/\mathfrak{P})^n$ [Cam18, SCL25] together with the standard Minkowski–Hlawka averaging argument over the base lattice $\sigma_K(\mathcal{O}_K^n)$ as in [CLB18, Sec. III]. Item (ii) follows from the same algebraic reduction argument used in [CLB18]: Dirichlet's unit theorem yields a full-rank lattice of logarithmic embeddings of units in $\mathcal{O}_K$, and a suitable unit gives the factorization $\tilde{H} = E_{\tilde{H}} U_{\tilde{H}}$ with bounded $\|E_{\tilde{H}}\|$.

Corollary 1 (Universal reliability at Bob). *Let $\{\mathbb{L}_n\}$ be the multiquadratic Construction π_A ensembles above, with fixed volume and increasing dimension N . Then there exists a sequence of lattices $\Lambda_b^{(n)} \in \mathbb{L}_n$ that is universally good for Bob's compound channel family S_b , in the sense of [CLB18].*

This is an immediate consequence of [CLB18, Theorem 1].

4.3 Secrecy via Flatness Factor and Algebraic Reduction

We now turn to secrecy. Intuitively, secrecy requires that Eve's received distribution be essentially independent of the transmitted message, so that different cosets become indistinguishable at the output.

Our goal is to obtain universal strong secrecy over Eve's compound set, namely

$$\sup_{H_e \in S_e} I(M; Y_e) \rightarrow 0 \quad \text{as } n \rightarrow \infty.$$

The secrecy argument follows [LLBS14, CLB20] and has three steps: (i) discrete Gaussian shaping randomizes the transmitted point inside each coset and reduces secrecy to a flatness-factor bound; (ii) algebraic reduction replaces the channel-dependent flatness factor by a spherical one at an equivalent variance σ_{eq} ; and (iii) the Minkowski–Hlawka property guarantees a sequence of lattices for which this spherical flatness factor vanishes.

We use a discrete Gaussian encoder over the coarse lattice Λ_e . For each message $s \in M_n$, Alice samples a lattice point in the coset $\Lambda_e + \lambda_s$ according to the discrete Gaussian distribution $D_{\Lambda_e + \lambda_s, \sigma_s}$ and transmits the corresponding codeword. Let $p_{Y_e|M=s}$ denote Eve's output density given $M = s$. If there exist a reference density q_{Y_e} and a sequence $\varepsilon_n \in [0, 1/4]$ such that $\mathbb{V}(p_{Y_e|M=s}, q_{Y_e}) \leq \varepsilon_n$ for all $s \in M_n$, then [CLB20, Lemma 2] implies

$$I(M; Y_e) \leq 8n\varepsilon_n R - 8\varepsilon_n \log_2(8\varepsilon_n).$$

Hence it is enough to prove that $\varepsilon_n = o(1/n)$ uniformly over $H_e \in S_e$.

Fix an admissible eavesdropper realization $H_e \in S_e$. As shown in [CLB20], the variational-distance parameter can be bounded by a flatness factor of the lattice $H_e \Lambda_e$ with respect to a correlated Gaussian

$$\varepsilon_n \leq \epsilon_{H_e \Lambda_e}(\sqrt{\Sigma_3}), \quad \Sigma_3^{-1} = (H_e H_e^\dagger)^{-1} \sigma_s^{-2} + \sigma_e^{-2} I.$$

The difficulty is that this quantity depends on the particular channel realization H_e . To obtain a bound that is uniform over the entire compound set, we introduce the worst-case quantity

$$C_e(H_e) := \log \det \left(I + \frac{\sigma_s^2}{\sigma_e^2} H_e H_e^\dagger \right), \quad C_e := \sup_{H_e \in S_e} C_e(H_e).$$

The next lemma replaces the channel-dependent flatness factor by a spherical one at a channel-independent equivalent variance.

Lemma 1 (Flatness factor under algebraic reduction, [CLB20]). *Assume that the dual lattice Λ^* admits algebraic reduction with constant $\alpha > 0$. Then, for every $H_e \in S_e$,*

$$\epsilon_{H_e \Lambda}(\sqrt{\Sigma_3}) \leq \epsilon_\Lambda(\sigma_{\text{eq}}), \quad \sigma_{\text{eq}}^2 = \sigma_s^2 \alpha^{-2} e^{-C_e/n_a},$$

where $\epsilon_\Lambda(\cdot)$ denotes the spherical flatness factor.

In our setting, the coarse lattice Λ_e is an algebraic Construction A lattice over $\mathcal{O}_K$ obtained from CRT codes. By Proposition 2, the corresponding ensemble satisfies the Minkowski–Hlawka property, and therefore we can choose a sequence $\{\Lambda_e^{(n)}\}$ such that $\epsilon_{\Lambda_e^{(n)}}(\sigma_{\text{eq}}) = o(1/n)$. By Lemma 1, this yields the uniform estimate

$$\sup_{H_e \in S_e} \epsilon_{H_e \Lambda_e^{(n)}}(\sqrt{\Sigma_3}) \leq \epsilon_{\Lambda_e^{(n)}}(\sigma_{\text{eq}}) = o(1/n),$$

and consequently $\sup_{H_e \in S_e} I(M; Y_e) \rightarrow 0$.

Theorem 1. *Let K be a multiquadratic field and let $\{(\Lambda_b^{(n)}, \Lambda_e^{(n)})\}_{n \geq 1}$ be nested lattices produced by multiquadratic Construction π_A , with $\Lambda_e^{(n)} \subset \Lambda_b^{(n)}$. Assume that: (i) the fine lattices $\Lambda_b^{(n)}$ are universally good for S_b in the sense of Corollary 1, and (ii) the coarse lattices $\Lambda_e^{(n)}$ are secrecy-good for AWGN at σ_{eq} , and their duals admit algebraic reduction with constant α . Then there exists a discrete Gaussian encoder based on $(\Lambda_b^{(n)}, \Lambda_e^{(n)})$ and a lattice decoder at Bob such that, for every $(H_b, H_e) \in S_b \times S_e$, Bob’s error probability and the leakage $I(M; Y_e)$ both vanish as $n \rightarrow \infty$, provided that the secrecy rate per channel use satisfies*

$$R < (C_b - C_e - n_a - 2n_a \log \alpha)^+.$$

Sketch of proof. Reliability follows from Corollary 1. For secrecy, choose $\Lambda_e^{(n)}$ so that $\epsilon_{\Lambda_e^{(n)}}(\sigma_{\text{eq}}) = o(1/n)$ using Proposition 2. Lemma 1 then gives a uniform bound over all $H_e \in S_e$, and substituting this into the variational-distance estimate above yields strong secrecy. The rate penalty $n_a + 2n_a \log \alpha$ is exactly the penalty inherited from the algebraic reduction step in [CLB20].

4.4 Illustrative Finite-Length Example

We illustrate the proposed construction on $K = \mathbb{Q}(\sqrt{17}, \sqrt{33})$, where the complete splitting of $\langle 2 \rangle$ yields four binary CRT levels. After embedding and zero-forcing equalization for a fixed 2×2 Rayleigh MIMO channel pair (H_b, H_e) , the decoder sees equivalent real channels $y_b = x + z_b$ and $y_e = x + z_e$, with $z_b \sim \mathcal{N}(0, I_N)$ and $z_e \sim \mathcal{N}(0, 6I_N)$, consistent with the single-block compound model. Each level uses an independent $(3, 6)$ -regular LDPC code of length $n = 800$ and rate $1/2$, generated by PEG and followed by a random interleaver. We apply multistage decoding with successive interference cancellation and layered belief propagation [HEA05, TV05, UHdL16]. Bob’s SNR is swept from 0 to 24 dB, and for each point we simulate until about 800 bit errors are collected at Bob.

The BER curves in Fig. 1 show a clear waterfall for Bob, while Eve remains at a much higher BER due to the effective noise-variance penalty $10 \log_{10} 6 \approx 7.8$ dB.

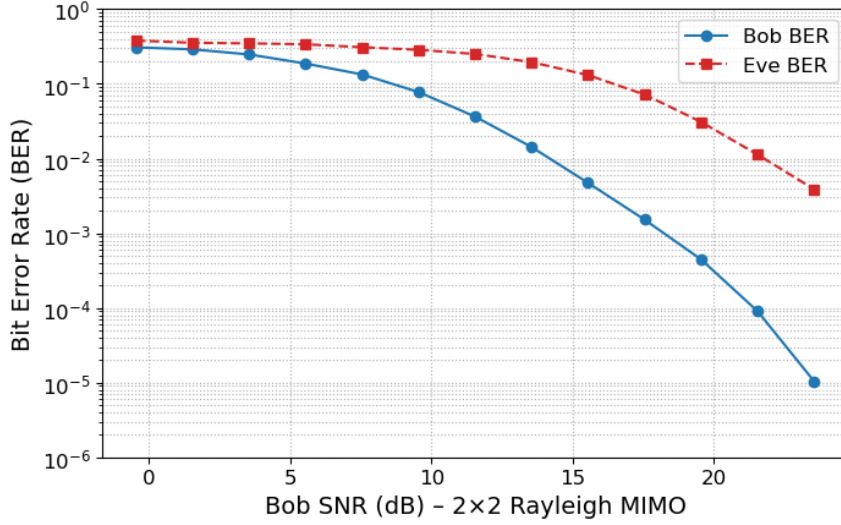


Figure 1: Bit-error rate versus Bob’s SNR for the Construction π_A over $K = \mathbb{Q}(\sqrt{17}, \sqrt{33})$ on a 2×2 Rayleigh MIMO wiretap channel.

5 Conclusions and Perspectives

We specialized Construction π_A over multiquadratic fields and obtained multilevel lattice codes over $\mathcal{O}_K$ that admit multistage decoding. When a rational prime splits completely, the induced CRT decomposition yields small residue alphabets, notably binary ones, so that standard LDPC component codes can be employed while preserving the algebraic structure of the construction. Placed within the algebraic Construction A framework, the resulting nested lattices apply to a compound block-fading wiretap channel, where discrete Gaussian shaping and flatness-factor bounds [LLBS14, CLB20] yield universal reliability for Bob and strong secrecy against Eve.

Our example over $K = \mathbb{Q}(\sqrt{17}, \sqrt{33})$ illustrates the use of binary component codes and the expected finite-length decoding behaviour. As future work, we plan to compare these multiquadratic Construction π_A ensembles with the multilevel algebraic partitions of [CLL17], and to develop finite-blocklength reliability and secrecy guarantees based on channel resolvability [PPV10, Hay06, BL13] for massive Internet-of-Things (IoT) and ultra-reliable low-latency communication (URLLC) scenarios with stringent energy and latency constraints [WYDP19].

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